The RFT Perspective Taking Protocol has very poor reliability and convergent validity with other measures of perspective taking, empathy, and theory of mind: A jingle assessment

Ian Hussey* & Malte Elson

Relational Frame Theory (RFT) conceptualises perspective taking and related constructs of empathy, theory of mind, and false-belief understanding in terms of deictic relational responding. RFT's Perspective Taking Protocol was originally developed to assess and train these relational repertoires, and increasingly became used to measure individual differences in perspective taking in adult and clinical samples. This slippage was recently raised as a concern regarding its appropriate use: the latter presupposes that PTP scores are reliable and valid measure of a perspective-taking trait, yet these properties have rarely been tested. This puts it at risk of jingle fallacy: the assumption similarly named measures measure similar things. We administered a 9-item PTP to 134 adults online, retesting 98 participants after a 10-day interval, alongside established self-report and behavioural measures of perspective taking, empathy, and theory of mind. The PTP showed poor internal consistency ($\alpha = .40$, 95% CI [.24, .54]) and test–retest reliability indistinguishable from zero (ICC2,1 = .08, 95% CI [−.12, .27]); its trial-type subscales did not cohere (subscale correlations from r = -.08 to .22); and it was essentially uncorrelated with existing measures even after disattenuating for unreliability. A meta-analysis demonstrated that the PTP is on average unrelated to other measures ($\bar{r} = .01$, disattenuated $\bar{r} = .03$) whereas those measures are generally associated with one another ($\bar{r} = .28$, disattenuated $\bar{r} = .35$). Results question whether the PTP should be used as a measure of individual differences in perspective taking; naming it a perspective-taking measure may represent a jingle fallacy.

Perspective taking, the related constructs of empathy, theory of mind, and false belief understanding, are important to many aspects of daily life and, as such, have received significant attention in the psychology literature (Hall & Schwartz, 2019). Relational Frame Theory (Hayes et al., 2001) conceptualises perspective taking, and indeed related phenomena such as false belief understanding, empathy and theory of mind in terms of deictic relational responding: interpersonal (I–you), spatial (here–there), and temporal (now–then; (McHugh et al., 2004). Perspective taking has received substantial empirical and conceptual attention within RFT research (for reviews, see Kavanagh et al., 2020; Montoya-Rodríguez et al., 2017).

The protocol originally developed by Barnes-Holmes (2001) to assess and train deictic relations

* Corresponding author: ian.hussey@unibe.ch

has seen a variety of modifications in subsequent work, and has been referred to by a variety of names including the deictic relations protocol (e.g., Kavanagh et al., 2020), the perspective taking protocol (e.g., McHugh et al., 2004), the deictic relational task (e.g., Walton et al., 2024), the Barnes-Holmes protocol (e.g., Hendriks et al., 2016), and possibly others. Throughout the current article, we refer to it as the perspective taking protocol (PTP) as this appears to be its most commonly used name.

On the premise that mature perspective-taking requires fluently abstracting and transforming relations across the self–other, here–there, and now–then dimensions, the PTP presents these through trials of escalating complexity: simple relations (e.g., "I have a red brick and you have a green brick; which brick do I have?"), reversed relations ("If I were you and you were me..."), and double-reversed relations that combine interpersonal and spatial/temporal reversals. RFT's account reframes perspective-taking as a learned repertoire that is potentially improvable through multiple exemplar training, a functional-analytic alternative to more mentalistic social-cognitive accounts (see Kavanagh et al., 2020). Across studies, several procedural variations have been made to the task, including changes to the complexity of the relational responses included (using some or all of simple, reversed, double reversed relations), the number of trials included, and abstract versus emotional content (see Supplementary Table S1). Importantly, for the sake of this article, its variations have also been used as both a training protocol (with corrective feedback, often in children) versus an assessment measure (with no corrective feedback, often in adults). This article concerns the latter variation of the task, which has been used for example to assess potential perspective taking deficits in social anhedonia (Vilardaga et al., 2012; Villatte et al., 2008), psychosis (Hendriks et al., 2016), intelligence and developmental delay (Gore et al., 2010), and Borderline Personality Disorder (Walton et al., 2024).

Psychometric Properties and Validity of the PTP

This use of the PTP as an assessment measure makes two, usually implicit, psychometric assumptions: that the sum-scores are 1) a reliable and 2) a valid measure of an individual differences trait. Yet, as Kavanagh et al. (2020) discuss, the protocol was explicitly designed as a training protocol to assess and establish deictic relations in children where they are absent, not to discriminate stably among adults who already possess this repertoire (see Kavanagh et al., 2020, for discussion of the changing use of the PTP over time). The properties required to establish the presence or absence of a repertoire are not those required to reliably rank-order individuals who generally possess it. Whether the PTP has these required properties for such usage has, to our knowledge, never been directly tested, despite its wide use in this manner (for review, see Montoya-Rodríguez et al., 2017).

Some concerns regarding the validity of the PTP arguably already exist. Hendriks et al. (2016) examined performance on a 62-trial version of the PTP found that it correlated only very modestly with two theory-of-mind tests across clinical (anxiety, psychosis) and healthy samples: correlations with the Faux-pas test (Baron-Cohen et al., 1999) were r = .32 to .42, and correlations with the Strange Stories Test (Happé, 1994) were r = .28 to .33. Notably, these correlational magnitudes are lower than would typically be accepted as evidence of convergent validity, and furthermore the correlations with the Strange Stories Test became non-significant when IQ was controlled for. This finding that the PTP is associated with IQ has been demonstrated elsewhere (Gore et al., 2010) and represents a strong risk of confounding to the PTP's interpretation.

Measurement Crisis in Psychology

This gap sits within a broader reckoning with measurement across psychology. Alongside the Replication Crisis (Munafò et al., 2017), a parallel Measurement Crisis has highlighted how rarely the validity of our measures is interrogated before they are used to draw conclusions (Flake & Fried, 2020; Lilienfeld & Strother, 2020; Schimmack, 2021). The consequences are not hypothetical: widely used scales harbour hidden invalidity that goes unnoticed because their psychometric properties are assumed rather than checked (Hussey & Hughes, 2020), and there is strong evidence that reliability metrics like Cronbach's alpha are subject to bias and hacking just as p-values are (Hussey et al., 2025).

Measurement practice also tends toward proliferation of unvalidated measures rather than consolidation towards well-validated ones (Anvari, Alsalti, Oehler, Hussey, et al., 2025; Anvari, Alsalti, Oehler, Marion, et al., 2025). Measures are treated like toothbrushes, with researchers preferring to invent their own rather than reuse others', multiplying the opportunities for the jingle fallacy: the assumption that measures that share similar names measure a similar construct (Elson et al., 2023; Thorndike, 1904).

Measurement concerns have already been raised for many measures and constructs used in this area. The widely used Reading the Mind in the Eyes Test (RMET: Baron-Cohen et al., 2001, 2015), a measure of theory of mind, has been shown to have very poor measurement properties, to the point that it has been argued it must be retired from use (Higgins et al., 2024, 2025; Higgins, Ross, et al., 2023; Higgins, Savalei, et al., 2023). Within RFT and Contextual Behavioral Science (Hayes et al., 2012) more broadly, the Implicit Relational Assessment Procedure has been shown to have very poor reliability, which bounds the correlations it can produce and greatly increases necessary sample sizes (Hussey & Drake, 2020). Measures of psychological flexibility show worryingly low item-content overlap between scales, raising jingle concerns (Ong et al., 2025), and mindfulness measures appear to be redundant with personality factors, a jangle fallacy (Altgassen et al., 2023).

Current Study

This study therefore tested the Perspective Taking Protocol's psychometric properties. Specifically, its internal consistency, its 10-day test-retest reliability, and its convergent validity with other well-established self-report and behavioral measures of perspective taking, theory of mind, and empathy. These broader constructs of theory of mind and empathy were included on the basis that conceptual RFT work frequently claims that deictic relational responding forms the basis of these other behavioral repertoires (see Kavanagh et al., 2020).

A 9-item version of the PTP was employed on the basis of item redundancy: in longer versions the items within each subscale are practically interchangeable and repeated, a carry-over from its original development as a training protocol where multiple exemplar training with corrective feedback was a design feature. In the context of psychometric assessment, these additional items do not expand construct coverage. The 9-item version contained three 'simple' trials, four 'reversed' trials, and two 'double reversed' trials, mirroring the overrepresentation of reversed trials in previous studies.

Method

Transparency statement

Full materials, scoring, analysis code, and the processed data are available at github.com/ianhussey/jingle-perspective-taking-task. The analyses were not preregistered.

Participants

Adults were recruited online via Prolific (prolific.co), with US or UK residence, native English and age ≥ 18 as eligibility criteria, and compensated £9 for completing both waves. Sample size was determined on the basis of resource availability and on the basis of an a priori precision power analysis that determined that 138 participants would provide 80% power to estimate test-retest reliability of at least $ICC_{2,1}$ = .50 to within ±0.20. As the precision of $ICC_{2,1}$ is a function of its magnitude, power and precision therefore increase at larger values (see Supplementary Materials). At Time 1, 134 participants provided usable data (2 excluded for failing embedded attention checks); 98 were retested at Time 2 after a 10-day interval (a further 2 excluded based on attention checks). The Time 1 sample averaged 38.2 years of age (SD = 13.0, range 19–75) and was 67.9% female, 31.3% male, and 0.7% non-binary. Data collection was conducted as part of a Bachelor's thesis project at the Faculty of Psychology at Ruhr University Bochum while both authors were employed there.

The Perspective-Taking Protocol

We administered a computer-based, forced-choice version of the McHugh et al. (2004) PTP implemented via Qualtrics. It consisted of 9 trials: 3 simple, 4 reversed, and 2 double-reversed. Each trial was scored correct (1) or incorrect (0); we computed a total score (0–9) and three trial-type subscale scores. The full list of items is available in the Supplementary Materials.

Comparison measures

Participants completed a battery of established self-report and behavioral measures spanning perspective taking, theory of mind, and empathy.

Self-reports. The Interpersonal Reactivity Index (IRI: Davis, 1983; 28 items), scored as both a total and four 7-item subscales (Perspective Taking, Fantasy, Empathic Concern, and Personal Distress). The Empathy Quotient (EQ: Baron-Cohen & Wheelwright, 2004; 40 scored items), scored as a total score and, following Lawrence et al. (2004), as cognitive-empathy, emotional-reactivity, and social-skills subscales. The Pictorial Empathy Test (PET: Lindeman et al., 2018; 7 items), as a measure of affective empathy. Lastly, the Single Item Trait Empathy Scale (SITES: Konrath et al., 2018), a single-item global empathy measure.

Behavioral tasks. The Reading the Mind in the Eyes Test (RMET; Baron-Cohen et al., 2001; 36 items), a test intended to measure theory of mind. Additionally, the Perspective-Taking Task for Adults (PTT-A: Brucato et al., 2023; 32 items), a behavioral measure of visual perspective taking. This was included on the basis of prior findings that spatial and emotional perspective taking may be related traits (Erle & Topolinski, 2015).

2.4 Analysis

Internal consistency was estimated as Cronbach's α (raw α with Feldt 95% CIs) on Time 1 data only. Test–retest reliability was estimated as the intraclass correlation coefficient, specifically two-way absolute agreement with single measurement ($ICC_{2,1}$), following Koo and Li (2016). Convergent validity was assessed using Pearson's r correlations among Time 1 scores. As a robustness test, we also employed Spearman's rho rank-order correlations, obtaining very similar results (see Supplementary Materials). To estimate the relationship between the measures' true (temporally stable) scores, we also disattenuated each correlation for measurement unreliability by dividing it by the square root of the product of the two measures' test–retest reliabilities: $r_{xy}^{corrected} = \frac{r_{xy}}{\sqrt{r_x r_y}}$. This also serves to control for reduction in reliability due to test shortening. Conclusions from these disattenuated correlations are therefore generalisable to longer versions of the task.

Finally, to summarise the convergent-validity pattern, we treated each (disattenuated) correlation as an effect size (Fisher r-to-z transformed) in a multi-level meta-analysis using the metafor R package (Viechtbauer, 2010). This model tested for differences (moderation) between the correlations that involve the PTP with those among non-PTP measures, after excluding within-instrument (part–whole) correlations. We employed cluster-robust standard errors and confidence intervals to account for the non-independence among the estimates. Pooled estimates are reported back-transformed to Pearson's r.

Figure 1. Raincloud plots of full-scale scores on the PTP and comparison measures, containing kernel density plots of distributions, box plots, and individual data points between the two time points.

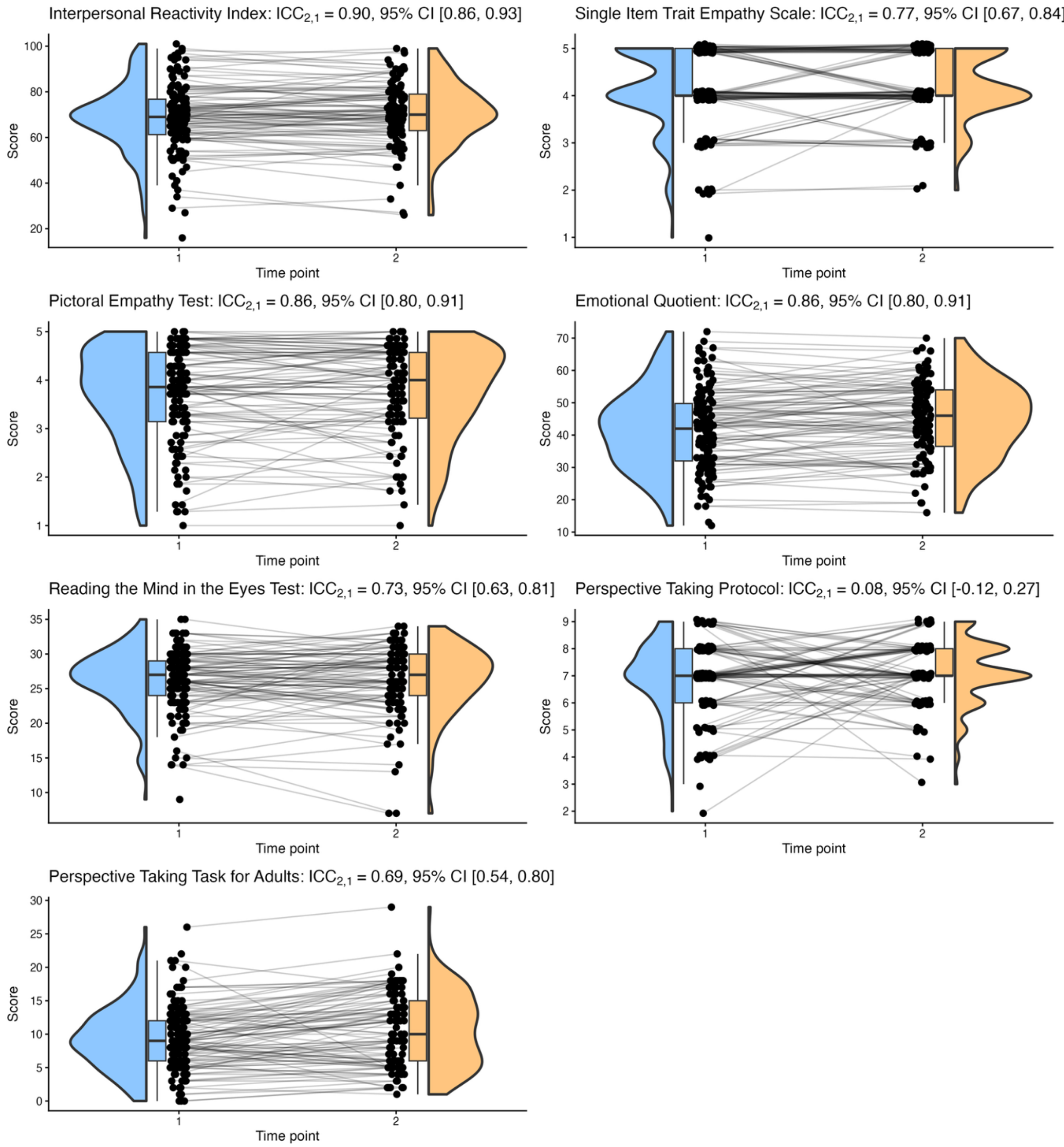
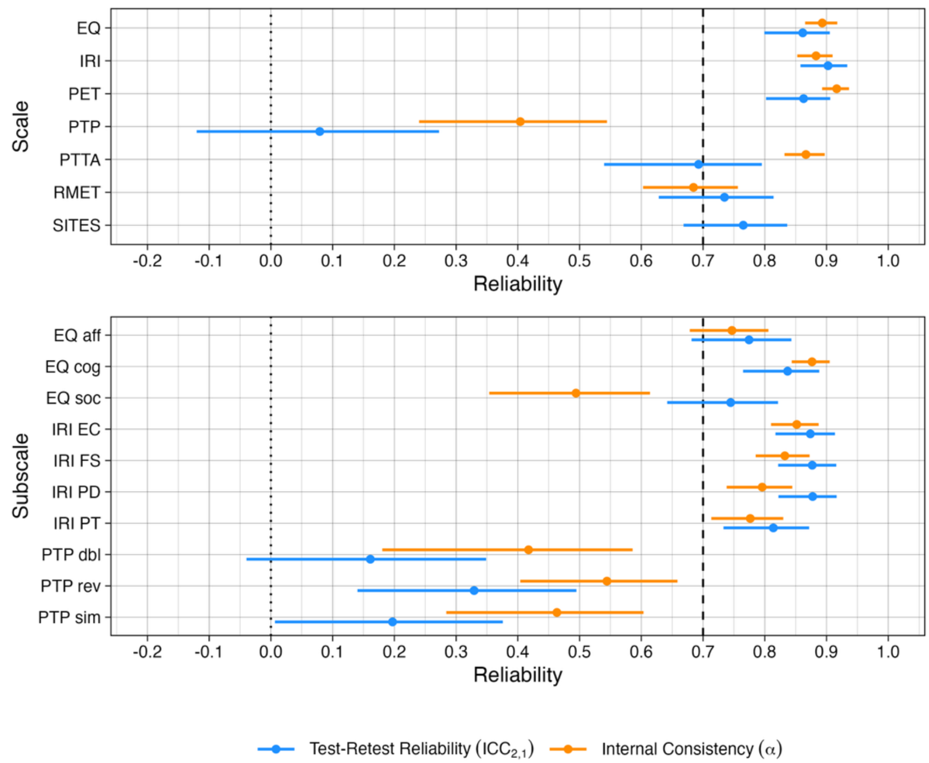


Figure 2. Reliability of the PTP (total and trial-type subscales) and comparison measures. Points are estimates and horizontal ranges are 95% confidence intervals; the dotted line marks zero, the dashed line marks a common acceptability threshold of .70. The PTP and its subscales (sim = simple, rev = reversed, dbl = double reversed) sit far below every other measure on both indices.



Results

Reliability

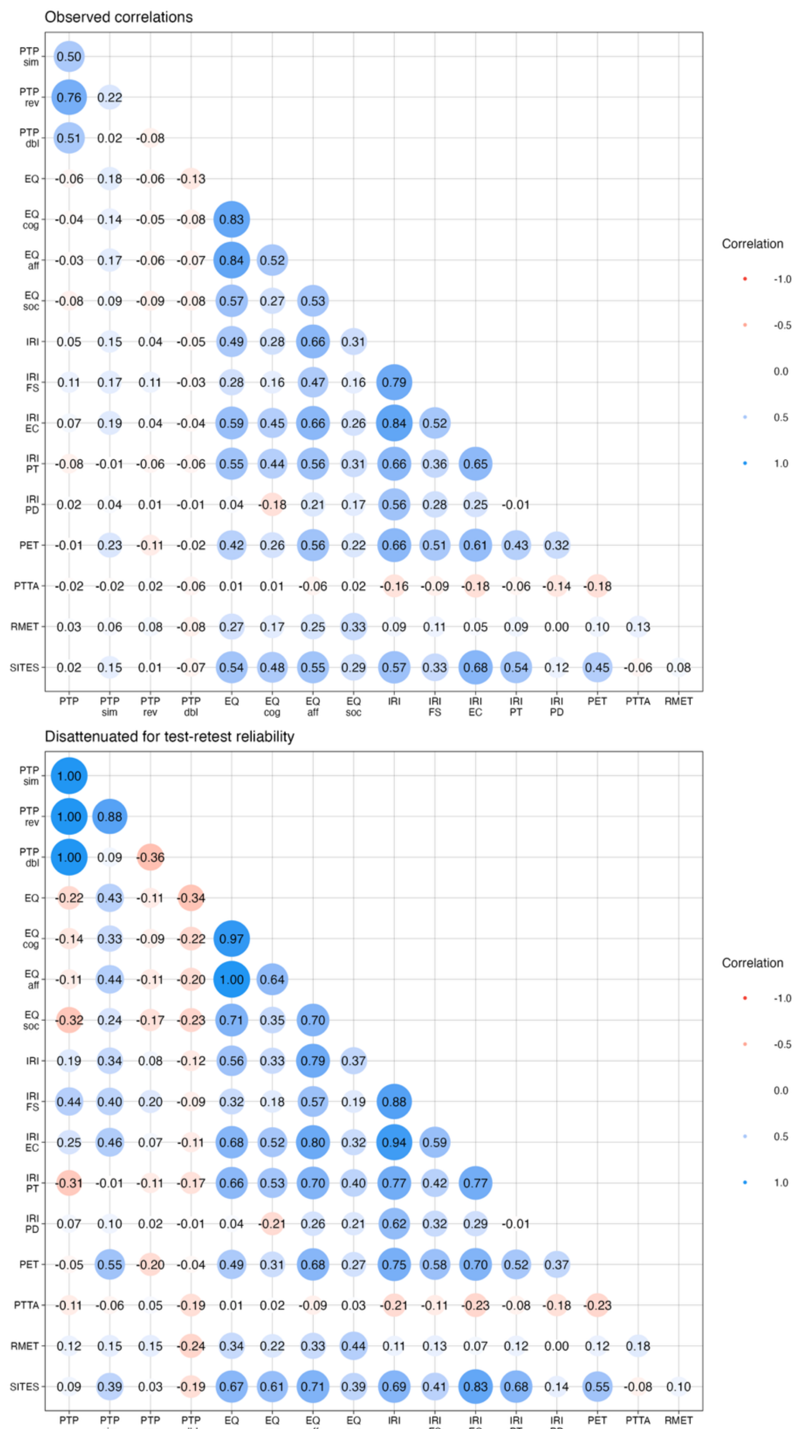
The PTP total score demonstrated poor internal consistency (α = .40) and very poor test–retest reliability that was not detectably different from zero ($ICC_{2,1}$ = .08, 95% CI [−.12, .27]). That is, a participant's PTP score at Time 1 carried essentially no detectable information about their score ten days later. All subscales demonstrated poor internal consistency (all α < .54), and very poor test-retest reliability (all $ICC_{2,1}$ < .35; see Figure 1 and Figure 2). In contrast, the comparison measures generally reached or were close to conventionally acceptable cut-offs on both indices (i.e., > .70; see Figure 2) with the exception of the EQ's social-skills subscale.

The PTP's subscales were found to be internally incoherent: its three trial-type subscales were mutually independent (simple–reversed r = .22; simple–double-reversed r = .02; reversed–double-reversed r = −.08), undermining the interpretability of its total score (see Figure 3).

Convergent validity

The PTP total score was generally uncorrelated with every other perspective-taking measure: IRI-PT (r = −.08), EQ-Cog (r = −.04), RMET (r = .03), and PTT-A (r = −.02). None approached significance (p = .37, .68, .75, and .78, respectively; see Figure 3). The same pattern held for the other comparison measures: no correlation exceeding | r | = .11 and none significant (all uncorrected $p \geq .18$; see Figure 3, upper panel).

Figure 3. Correlations among all (sub)scales at Time 1: observed Pearson correlations (upper panel); correlations disattenuated for each pair's test–retest reliability (lower panel). PTP subscales: sim = simple, rev = reversed, dbl = double reversed. Disattenuated values are clamped to [−1, 1] for display; out-of-range cells (the PTP part–whole pairs, inflated by the PTP's near-zero reliability) are reported exactly in the tables in the Supplementary Materials.



Disattenuating these correlations for measurement unreliability did not change this conclusion. Correcting each PTP–measure correlation by the test–retest reliabilities of the two measures revealed small or negligible correlations: IRI-PT $r = -.31$, EQ-Cog $r = -.14$, RMET $r = .12$, and PTT-A $r = -.11$, the opposite direction to that expected. The more robust reading is simply that there is too little reliable PTP variance for any true correlation to be recovered. By the same correction, the comparison measures generally converged, with the exception of the RMET (which has existing concerns about its validity) and the PTT-A which, as a behavioral measure, may be at risk of hidden invalidity (e.g., Buelow et al., 2024; Frey et al., 2017; Külpmann et al., 2026; Lilienfeld & Strother, 2020; Parsons et al., 2019). See Figure 3 (lower panel).

This contrast was formalised and tested using a moderator meta-analysis with robust cluster standard errors to acknowledge the non-independence of the correlations. Across the $k = 114$ cross-instrument correlations (52 involving the PTP, 62 among non-PTP measures, excluding within-scale/subscale correlations), correlations involving the PTP or its subscales with the contrast measures pooled at an average $\bar{r} = .01$, 95% CI $[-.01, .04]$, compared to correlations among the contrast measures having an average $\bar{r} = .28$ [.20, .36] (moderator $p < .001$). Applying the same analysis to the correlations after disattenuating for test–retest reliability did not narrow the gap: PTP-contrast correlations $\bar{r} = .03$ $[-.05, .11]$ versus contrast-contrast correlations $\bar{r} = .35$ [.24, .46] (moderator $p < .001$). Because the RMET itself has poor and contested psychometric properties (Higgins et al., 2024, 2025; Higgins, Ross, et al., 2023; Higgins, Savalei, et al., 2023), as a sensitivity analysis we repeated the meta-analysis excluding every correlation involving it ($k = 98$). Very similar results were observed: contrast-contrast correlations $\bar{r} = .31$ [.22, .40] and disattenuated correlations $\bar{r} = .39$ [.27, .50], versus PTP-contrast correlations $\bar{r} = .01$ $[-.02, .04]$ and disattenuated correlations $\bar{r} = .03$ $[-.05, .11]$; both moderator ps $< .001$. We did not exclude the PTT-A, as no concerns about its validity had been raised prior to this study.

Discussion

A measure used to rank people on perspective taking should rank them consistently and should agree with other measures of the same or similar constructs. The shortened PTP did neither: poor internal consistency, test–retest reliability near zero, trial-type subscales that do not cohere, and no or very limited shared variance with four other accepted measures of perspective-taking. While one could argue post hoc that deictic relational responding differs meaningfully from perspective taking, empathy, and theory of mind in their social-cognitive sense, existing RFT theorising has clearly argued that it is closely related to these abilities and should therefore correlate with them (Kavanagh et al., 2020). Additionally, when reliability is this low it is not clear the task can measure any trait validly, since some degree of reliability is a mathematical prerequisite for validity (for a worked example using the IRAP, see Hussey, 2025). The most parsimonious reading is that PTP scores are dominated by occasion-specific error rather than any stable property of the person, and that whatever residual signal they contain is unrelated to what other "perspective-taking" or other measures capture. Using the name "perspective taking" for this measure is, on the present evidence, therefore a jingle fallacy.

This is not a wholesale indictment of the RFT account of perspective taking, nor of the deictic-relations analysis. The current results are agnostic to the use of the PTP as a training protocol to establish such repertoires. Rather, it is an indictment of treating a developmental assessment-and-training protocol as an individual-differences measure, without assessment of its measurement properties. This is precisely the slippage flagged by Kavanagh et al. (2020).

The PTP is not the only widely used measure this study raises concerns about. The Reading the Mind in the Eyes Test (RMET), included here as a theory-of-mind benchmark and itself converging only weakly with the other measures (all $|r| \leq .17$) has, since this study was run in 2023, also been examined in a similar way with similarly poor results, resulting in the recommendation that the test be retired from use (Higgins et al., 2024, 2025; Higgins, Ross, et al., 2023; Higgins, Savalei, et al., 2023). In the current study's data, the PTP was found to be even less reliable than the RMET (α = .40 vs. .68; $ICC_{2,1}$ = .08 vs. .73) while sharing the same near-zero convergent validity. If the evidentiary standard that warrants retiring the RMET is the right one, the PTP should also be retired from use as a measure of individual differences.

Lastly, in contrast with prior results by Erle and Topolinski (2015), spatial perspective taking on the PTT-A was not found to be associated with other forms of emotional perspective taking, representing a conceptual replication failure.

Limitations

We tested a shortened version of the PTP. A longer protocol with more trials per relation would, following Classical Test Theory, improve reliability, but is unlikely to improve the convergent-validity null results. The robustness of the results to disattenuation for reliability suggests that this lack of convergent validity is likely to generalise to longer versions of the task. The sample was limited to a non-clinical, native-English online panel. Attention checks served to ensure data quality, but the generalisability of the results to clinical populations would benefit from future examination.

Conclusion

The Perspective-Taking Protocol appears to suffer from the jingle fallacy: naming it a measure of perspective taking does not make it one. It was extremely unreliable, given its internal consistency is low and its test–retest reliability near zero, and had near-zero convergent validity with other measures of perspective taking, theory of mind, or empathy. A longer version would likely be more reliable, given that reliability is a function of task length assuming parallel effects (e.g., absence of fatigue effects etc.); however, increasing task length alone is unlikely to produce a reliable and valid measure given item reptition in longer versions.

Author note

Ian Hussey https://orcid.org/0000-0001-8906-7559

Malte Elson https://orcid.org/0000-0001-7806-9583

Correspondence regarding this article should be addressed to Ian Hussey, Institute of Psychology, Fabrikstrasse 8, University of Bern, Bern 3011, Switzerland. Email: ian.hussey@unibe.ch.

We would like to thank Markus Dieterich who assisted with the study design and data collection.

References

Altgassen, E., Geiger, M., & Wilhelm, O. (2023). Do you mind a closer look? A jingle-jangle fallacy perspective on mindfulness. *European Journal of Personality*, 08902070231174575. https://doi.org/10.1177/08902070231174575

Anvari, F., Alsalti, T., Oehler, L. A., Hussey, I., Elson, M., & Arslan, R. C. (2025). Defragmenting psychology. *Nature Human Behaviour*, 1–4. https://doi.org/10.1038/s41562-025-02138-0

Anvari, F., Alsalti, T., Oehler, L. A., Marion, Z., Hussey, I., Elson, M., & Arslan, R. C. (2025). A Fragmented Field: Construct and Measure Proliferation in Psychology. *Advances in Methods and Practices in Psychological Science*. (Sage CA: Los Angeles, CA). https://journals.sagepub.com/doi/10.1177/25152459251360642

Baron-Cohen, S., Bowen, D. C., Holt, R. J., Allison, C., Auyeung, B., Lombardo, M. V., Smith, P., & Lai, M.-C. (2015). The "Reading the Mind in the Eyes" Test: Complete Absence of Typical Sex Difference in ~400 Men and Women with Autism. *PLOS ONE*, *10*(8), e0136521. https://doi.org/10.1371/journal.pone.0136521

Baron-Cohen, S., O'Riordan, M., Stone, V., Jones, R., & Plaisted, K. (1999). Recognition of Faux Pas by Normally Developing Children and Children with Asperger Syndrome or High-Functioning Autism. *Journal of Autism and Developmental Disorders*, *29*(5), 407–418. https://doi.org/10.1023/A:1023035012436

Baron-Cohen, S., & Wheelwright, S. (2004). The empathy quotient: An investigation of adults with Asperger syndrome or high functioning autism, and normal sex differences. *Journal of Autism and Developmental Disorders*, *34*(2), 163–175. https://doi.org/10.1023/b:jadd.0000022607.19833.00

Baron-Cohen, S., Wheelwright, S., Hill, J., Raste, Y., & Plumb, I. (2001). The "Reading the

Mind in the Eyes" Test Revised Version: A Study with Normal Adults, and Adults with Asperger Syndrome or High-functioning Autism. *Journal of Child Psychology and Psychiatry*, *42*(2), 241–251. https://doi.org/10.1111/1469-7610.00715

Brucato, M., Frick, A., Pichelmann, S., Nazareth, A., & Newcombe, N. S. (2023). Measuring Spatial Perspective Taking: Analysis of Four Measures Using Item Response Theory. *Topics in Cognitive Science*, *15*(1), 46–74. https://doi.org/10.1111/tops.12597

Buelow, M. T., Kowalsky, J. M., & Okdie, B. M. (2024). Test–Retest Reliability of Common Behavioral Decision-Making Tasks: A Multi-Sample, Repeated Measures Study. *Archives of Clinical Neuropsychology*, *39*(3), 378–382.

Davis, M. H. (1983). Measuring individual differences in empathy: Evidence for a multidimensional approach. *Journal of Personality and Social Psychology*, *44*, 113–126. https://doi.org/10.1037/0022-3514.44.1.113

Elson, M., Hussey, I., Alsalti, T., & Arslan, R. C. (2023). Psychological measures aren't toothbrushes. *Communications Psychology*, *1*(1), Article 1. https://doi.org/10.1038/s44271-023-00026-9

Erle, T. M., & Topolinski, S. (2015). Spatial and Empathic Perspective-Taking Correlate on a Dispositional Level. *Social Cognition*, *33*(3), 187–210. https://doi.org/10.1521/soco.2015.33.3.187

Flake, J. K., & Fried, E. I. (2020). Measurement Schmeasurement: Questionable Measurement Practices and How to Avoid Them. *Advances in Methods and Practices in Psychological Science*, *3*(4), 456–465. https://doi.org/10.1177/2515245920952393

Frey, R., Pedroni, A., Mata, R., Rieskamp, J., & Hertwig, R. (2017). Risk preference shares the psychometric structure of major psychological traits. *Science Advances*, *3*(10), e1701381. https://doi.org/10.1126/sciadv.1701381

Gore, N. G., Barnes-Holmes, Y., & Murphy, G. (2010). The Relationship between Intellectual Functioning and Relational Perspective-Taking. *International Journal of Psychology and Psychological Therapy*, *10*(1), 1–17.

Hall, J. A., & Schwartz, R. (2019). Empathy present and future. *The Journal of Social Psychology*, *159*(3), 225–243. https://doi.org/10.1080/00224545.2018.1477442

Happé, F. G. E. (1994). An advanced test of theory of mind: Understanding of story characters' thoughts and feelings by able autistic, mentally handicapped, and normal children and adults. *Journal of Autism and Developmental Disorders*, *24*(2), 129–154. https://doi.org/10.1007/BF02172093

Hayes, S. C., Barnes-Holmes, D., & Roche, B. (2001). *Relational frame theory: A post-Skinnerian account of human language and cognition*. Kluwer Academic/Plenum Press.

Hayes, S. C., Barnes-Holmes, D., & Wilson, K. G. (2012). Contextual Behavioral Science: Creating a science more adequate to the challenge of the human condition. *Journal of Contextual Behavioral Science*, *1*(1–2), 1–16. https://doi.org/10.1016/j.jcbs.2012.09.004

Hendriks, A. L., Barnes-Holmes, Y., McEnteggart, C., De Mey, H., Witteman, C., Janssen, G., & Egger, J. (2016). The relationship between theory of mind and relational frame theory: Convergence of perspective-taking measures. *Clinical Neuropsychiatry*, *13*, 17–23.

Higgins, W. C., Kaplan, D. M., Deschrijver, E., & Ross, R. M. (2024). Construct validity evidence reporting practices for the Reading the Mind in the Eyes Test: A systematic scoping review. *Clinical Psychology Review*, *108*, 102378. https://doi.org/10.1016/j.cpr.2023.102378

Higgins, W. C., Kaplan, D. M., Deschrijver, E., & Ross, R. M. (2025). Why most research based on the Reading the Mind in the Eyes Test is unsubstantiated and uninterpretable: A response to Murphy and Hall (2024). *Clinical Psychology Review*, *115*, 102530. https://doi.org/10.1016/j.cpr.2024.102530

Higgins, W. C., Ross, R. M., Langdon, R., & Polito, V. (2023). The "Reading the Mind in the Eyes" Test Shows Poor Psychometric Properties in a Large, Demographically Representative U.S. Sample. *Assessment*, *30*(6), 1777–1789. https://doi.org/10.1177/10731911221124342

Higgins, W. C., Savalei, V., Polito, V., & Ross, R. M. (2023). Validation of the reading the mind in the eyes test requires an interpretable factor model. *Proceedings of the National Academy of Sciences*, *120*(52), e2303706120. https://doi.org/10.1073/pnas.2303706120

Hussey, I. (2025). Verification report: A critical reanalysis of Vahey et al. (2015) "A meta-analysis of criterion effects for the Implicit Relational Assessment Procedure (IRAP) in the clinical domain". *Journal of Behavior Therapy and Experimental Psychiatry*, 102015. https://doi.org/10.1016/j.jbtep.2024.102015

Hussey, I., Alsalti, T., Bosco, F., Elson, M., & Arslan, R. (2025). An Aberrant Abundance of Cronbach's Alpha Values at .70. *Advances in Methods and Practices in Psychological Science*, *8*(1), 25152459241287123. https://doi.org/10.1177/25152459241287123

Hussey, I., & Drake, C. E. (2020). The Implicit Relational Assessment Procedure demonstrates poor internal consistency and test-retest reliability: A meta-analysis. *PsyArXiv*. https://doi.org/10.31234/osf.io/ge3k7

Hussey, I., & Hughes, S. (2020). Hidden Invalidity Among 15 Commonly Used Measures in Social and Personality Psychology. *Advances in Methods and Practices in Psychological Science*, *3*(2), 166–184. https://doi.org/10.1177/2515245919882903

Kavanagh, D., Barnes-Holmes, Y., & Barnes-Holmes, D. (2020). The Study of Perspective-Taking: Contributions from Mainstream Psychology and Behavior Analysis. *The Psychological Record*, *70*(4), 581–604. https://doi.org/10.1007/s40732-019-00356-3

Konrath, S., Meier, B. P., & Bushman, B. J. (2018). Development and validation of the single item trait empathy scale (SITES). *Journal of Research in Personality*, *73*, 111–122. https://doi.org/10.1016/j.jrp.2017.11.009

Koo, T. K., & Li, M. Y. (2016). A Guideline of Selecting and Reporting Intraclass Correlation Coefficients for Reliability Research. *Journal of Chiropractic Medicine*, *15*(2), 155–163. https://doi.org/10.1016/j.jcm.2016.02.012

Külpmann, A. I., Ries, J.-P., Hussey, I., & Elson, M. (2026). *Methodological Flexibility in the Iowa Gambling Task Undermines Interpretability: A Meta-method Review* (4g3vr_v1). PsyArXiv. https://osf.io/preprints/psyarxiv/4g3vr_v1/

Lawrence, E. J., Shaw, P., Baker, D., Baron-Cohen, S., & David, A. S. (2004). Measuring empathy: Reliability and validity of the Empathy Quotient. *Psychological Medicine*, *34*(5), 911–920. https://doi.org/10.1017/S0033291703001624

Lilienfeld, S. O., & Strother, A. N. (2020). Psychological measurement and the replication crisis: Four sacred cows. *Canadian Psychology / Psychologie Canadienne*, *61*, 281–288. https://doi.org/10.1037/cap0000236

Lindeman, M., Koirikivi, I., & Lipsanen, J. (2018). Pictorial Empathy Test (PET). *European Journal of Psychological Assessment*, *34*(6), 421–431. https://doi.org/10.1027/1015-5759/a000353

McHugh, L., Barnes-Holmes, Y., & Barnes-Holmes, D. (2004). Perspective-taking as relational responding: A developmental profile. *Psychological Record*, *54*(1), 115–144. Scopus.

Montoya-Rodríguez, M. M., Molina, F. J., & McHugh, L. (2017). A Review of Relational Frame Theory Research Into Deictic Relational Responding. *The Psychological Record*, *67*(4), 569–579. https://doi.org/10.1007/s40732-016-0216-x

Munafò, M. R., Nosek, B. A., Bishop, D. V. M., Button, K. S., Chambers, C. D., Du Sert, N. P., Simonsohn, U., Wagenmakers, E.-J., Ware, J. J., & Ioannidis, J. P. A. (2017). A manifesto for reproducible science. *Nature Human Behaviour*, *1*(1), 0021. https://doi.org/10.1038/s41562-016-0021

Ong, C. W., Skolnik, A. M., Johnson, H. M., & Lee, E. B. (2025). Examining item content across nine psychological (in)flexibility scales: What do they measure? *Journal of Contextual Behavioral Science*, *35*, 100872. https://doi.org/10.1016/j.jcbs.2025.100872

Parsons, S., Kruijt, A.-W., & Fox, E. (2019). Psychological Science Needs a Standard Practice of Reporting the Reliability of Cognitive-Behavioral Measurements. *Advances in Methods and Practices in Psychological Science*, *2*(4), 378–395. https://doi.org/10.1177/2515245919879695

Schimmack, U. (2021). The Validation Crisis in Psychology. *Meta-Psychology*, *5*. https://doi.org/10.15626/MP.2019.1645

Thorndike, E. L. (1904). *Theory of mental and social measurements* (pp. xii, 212). The Science Press. https://doi.org/10.1037/13283-000

Viechtbauer, W. (2010). Conducting Meta-Analyses in R with the metafor Package. *Journal of Statistical Software*, *36*(3). https://doi.org/10.18637/jss.v036.i03

Vilardaga, R., Estévez, A., Levin, M. E., & Hayes, S. C. (2012). Deictic Relational Responding, Empathy, and Experiential Avoidance as Predictors of Social Anhedonia: Further Contributions From Relational Frame Theory. *The Psychological Record*, *62*(3), 409–432. https://doi.org/10.1007/BF03395811

Villatte, M., Monestès, J.-L., McHugh, L., Freixa i Baqué, E., & Loas, G. (2008). Assessing deictic relational responding in social anhedonia: A functional approach to the development of theory of mind impairments. *International Journal of Behavioral Consultation and Therapy*, *4*(4), 360–373. https://doi.org/10.1037/h0100867

Walton, C. J., Rasmussen, A., Villatte, M., Viera, R. V., Irwin, L., & Rossiter, R. (2024). A relational frame approach to perspective taking in persons with Borderline Personality Disorder. *Journal of Contextual Behavioral Science*, *32*, 100777. https://doi.org/10.1016/j.jcbs.2024.100777